

Exact Quarter-Base Extrapolation for Dyadic Samples of Rvachev's Up-Function Spline Prefixes

A technical note accompanying the ProveIt Fabius-function development

2 September 2026

Abstract

Let

$$\widehat{\text{up}}(t) = \prod_{k=0}^{\infty} \text{sinc}\left(\frac{\pi t}{2^k}\right), \quad \text{sinc } z = \frac{\sin z}{z},$$

and let up_n be the inverse Fourier transform of the prefix through $k = n$. For a fixed dyadic $x \in [-1, 1]$, put $q_n = \text{up}_n(x)$. We prove that after the spline grid resolves x , the sequence is not merely asymptotic to a finite sum of exponentials: it is *exactly* of the form

$$q_n = \text{up}(x) + \sum_{r=1}^d A_r(x) 4^{-rn}, \quad n \geq s,$$

where $a = 1 - |x| = m/2^s$ is reduced and $d = \lfloor s/2 \rfloor$. For every nontrivial depth $s \geq 2$, all $A_1(x), \dots, A_d(x)$ are nonzero, so this is the exact number of geometric modes.

The constant term can consequently be recovered from only $d + 1$ consecutive spline values. With $\rho = 1/4$, any $N \geq s$ gives the single closed formula

$$\text{up}(x) = \frac{1}{(\rho; \rho)_d} \sum_{j=0}^d (-1)^j \rho^{\binom{j+1}{2}} \begin{bmatrix} d \\ j \end{bmatrix}_{\rho} q_{N+d-j}.$$

This is a quarter-base Gaussian-binomial/Richardson row. Its generating polynomial is $(\rho z; \rho)_d / (\rho; \rho)_d$, whose roots annihilate all the decaying modes. We also give the exact Vandermonde system for all coefficients, a triangular extraction algorithm, stability bounds, and exact rational examples. The accompanying Python program uses only the standard library and verifies every displayed numerical identity with `fractions.Fraction` arithmetic.

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1 Statement of the problem and the answer

We use the Fourier convention

$$\widehat{f}(t) = \int_{\mathbb{R}} f(x) e^{-2\pi i x t} dx, \quad f(x) = \int_{\mathbb{R}} \widehat{f}(t) e^{2\pi i x t} dt. \quad (1)$$

The Rvachev up-function is the even, nonnegative, compactly supported C^∞ function normalized by

$$\widehat{\text{up}}(t) = \prod_{k=0}^{\infty} \text{sinc}\left(\frac{\pi t}{2^k}\right). \quad (2)$$

For $n \geq 0$, define the finite-product approximation by

$$\widehat{\text{up}}_n(t) = \prod_{k=0}^n \text{sinc}\left(\frac{\pi t}{2^k}\right). \quad (3)$$

There are $n + 1$ sinc factors in (3); this indexing point matters. The inverse transform up_n is a compactly supported piecewise polynomial of degree n .

Fix a dyadic $x \in [-1, 1]$ and put

$$q_n := \text{up}_n(x), \quad a := 1 - |x| \in [0, 1]. \quad (4)$$

Write a in reduced dyadic form

$$a = \frac{m}{2^s}, \quad 0 \leq m \leq 2^s, \quad (5)$$

where $s = 0$ for $a \in \{0, 1\}$ and, otherwise, m is odd. We call s the reduced dyadic depth of a and set

$$d := \left\lfloor \frac{s}{2} \right\rfloor, \quad \rho := \frac{1}{4}. \quad (6)$$

The main conclusions are the following.

Theorem 1.1 (Exact eventual geometric tail). *For every dyadic $x \in [-1, 1]$, with s, d as above, there are rational coefficients $A_1(x), \dots, A_d(x)$ such that*

$$q_n = \text{up}(x) + \sum_{k=1}^d A_k(x) \rho^{kn} \quad (n \geq s). \quad (7)$$

At the three depth-zero points, the statement is understood directly and is valid already at $n = 0$. If $s \geq 2$, then every displayed coefficient $A_k(x)$ is nonzero; hence the number d of decaying modes is exact.

Theorem 1.2 (Single q-binomial extraction formula). *Let $N \geq s$. Then*

$$\text{up}(x) = \frac{1}{(\rho; \rho)_d} \sum_{j=0}^d (-1)^j \rho^{\binom{j+1}{2}} \left[\begin{matrix} d \\ j \end{matrix} \right]_{\rho} q_{N+d-j}, \quad \rho = \frac{1}{4}. \quad (8)$$

Here

$$(z; q)_d = \prod_{r=0}^{d-1} (1 - zq^r), \quad \left[\begin{matrix} d \\ j \end{matrix} \right]_q = \frac{(q; q)_d}{(q; q)_j (q; q)_{d-j}}. \quad (9)$$

In particular, the canonical choice $N = s$ recovers the exact rational value from $q_s, q_{s+1}, \dots, q_{s+d}$.

The rest of the article proves these statements, identifies the coefficients, and turns the closed formula into an exact algorithm.

2 Finite sinc products are centered Fabius splines

2.1 Box densities and a random-sum interpretation

For $k \geq 0$, put

$$b_k(x) = 2^k \mathbf{1}_{[-2^{-k-1}, 2^{-k-1}]}(x). \quad (10)$$

A direct integration using (1) gives

$$\widehat{b}_k(t) = \text{sinc}\left(\frac{\pi t}{2^k}\right). \quad (11)$$

Consequently

$$\text{up}_n = b_0 * b_1 * \dots * b_n. \quad (12)$$

Equivalently, up_n is the probability density of

$$Z_n = \sum_{k=0}^n 2^{-k} V_k, \quad V_k \sim \text{Unif}\left[-\frac{1}{2}, \frac{1}{2}\right] \quad \text{independently}. \quad (13)$$

This immediately proves evenness and nonnegativity. It also shows that the support is

$$\text{supp up}_n = \left[-1 + 2^{-n-1}, 1 - 2^{-n-1}\right]. \quad (14)$$

2.2 The explicit Thue–Morse spline

Let

$$\varepsilon_r = (-1)^{s_2(r)}, \quad (15)$$

where $s_2(r)$ is the number of 1's in the binary expansion of r . For $n \geq 1$ define the centered spline

$$S_n(y) = \frac{1}{2^{\binom{n}{2}} n!} \sum_{r \geq 0} \varepsilon_r \left(2^n y - \frac{1}{2} - r\right)_+^n. \quad (16)$$

At a fixed y , the sum is finite. A convenient cutoff is

$$N_n(y) = \max\left(0, \left\lfloor 2^n y + \frac{1}{2} \right\rfloor\right), \quad 0 \leq r < N_n(y). \quad (17)$$

Lemma 2.1 (Exact indexing bridge). *For every $n \geq 1$ and every $x \in [-1, 1]$,*

$$\boxed{\text{up}_n(x) = S_n(1+x)}. \quad (18)$$

On the same interval, evenness also gives

$$\text{up}_n(x) = S_n(1 - |x|). \quad (19)$$

Proof. The standard truncated-power formula for the density of a sum of independent uniform variables, applied to the widths 2^{-k} for $0 \leq k \leq n$, gives

$$\text{up}_n(x) = \frac{1}{n! \prod_{k=0}^n 2^{-k}} \sum_{A \subseteq \{0, \dots, n\}} (-1)^{|A|} \left(x + \frac{1}{2} \sum_{k=0}^n 2^{-k} - \sum_{k \in A} 2^{-k} \right)_+^n. \quad (20)$$

Encode a subset A by

$$r = \sum_{k \in A} 2^{n-k}. \quad (21)$$

Then r runs from 0 to $2^{n+1} - 1$ and $(-1)^{|A|} = \varepsilon_r$. Since

$$\frac{1}{2} \sum_{k=0}^n 2^{-k} = 1 - 2^{-n-1}, \quad \prod_{k=0}^n 2^{-k} = 2^{-n(n+1)/2}, \quad (22)$$

For $x \in [-1, 1]$, no index beyond $2^{n+1} - 1$ can contribute. Hence rescaling the positive part in (20) yields

$$\text{up}_n(x) = \frac{1}{2^{\binom{n}{2}} n!} \sum_{r \geq 0} \varepsilon_r \left(2^n(1+x) - \frac{1}{2} - r \right)_+^n, \quad (23)$$

which is (18). Equation (19) follows because every factor b_k in (12) is even. $\square$

Remark 2.2 (The indexing checkpoint). *Some treatments use a prefix with factors $k = 0, \dots, n-1$. That convention produces S_{n-1} rather than S_n . In this note, the user's product ends at $k = n$, contains $n+1$ factors, and produces the degree- n spline S_n .*

2.3 The Fabius distribution and the fold to up

Let $U_1, U_2, \dots$ be independent uniform random variables on $[0, 1]$ and put

$$Y = \sum_{j=1}^{\infty} \frac{U_j}{2^j}. \quad (24)$$

Let F be the distribution function of Y . Then F is the bounded Fabius function [1, 3, 6], and

$$Z = 2Y - 1 \quad (25)$$

has density up on $[-1, 1]$. The familiar fold relation is

$$\text{up}(x) = F(1 - |x|), \quad -1 \leq x \leq 1. \quad (26)$$

For

$$Y_n = \sum_{j=1}^n \frac{U_j}{2^j}, \quad X_n = Y_n + 2^{-n-1}, \quad (27)$$

the spline in (16) is exactly the CDF of X_n :

$$S_n(y) = \mathbb{P}(X_n \leq y), \quad 0 \leq y \leq 1. \quad (28)$$

This follows either from the same uniform-sum truncated-power formula or by induction from the smoothing recurrence.

Combining (4), (19), and (26), our problem is therefore equivalent to this one:

$$q_n = S_n(a) \longrightarrow F(a), \quad a = \frac{m}{2^s} \in [0, 1]. \quad (29)$$

3 Exact matched-cell moment deconvolution

The decisive observation is that the omitted random tail has exactly the half-width of one spline cell.

3.1 The centered tail and its moments

Let Y' be an independent copy of Y . The tail after n binary scales is

$$Y - Y_n \stackrel{d}{=} 2^{-n} Y'. \quad (30)$$

Consequently, with $Z' = 2Y' - 1$ and

$$\delta_n := 2^{-n-1}, \quad (31)$$

we have the centered decomposition

$$Y \stackrel{d}{=} X_n + \delta_n Z'. \quad (32)$$

Conditioning on Z' gives the exact convolution identity

$$F(t) = \mathbb{E}[S_n(t - \delta_n Z)] = \int_{-1}^1 S_n(t - \delta_n z) \text{up}(z) \, dz. \quad (33)$$

Define the even moments

$$c_k := \mathbb{E}(Z^{2k}) = \int_{-1}^1 z^{2k} \text{up}(z) \, dz, \quad c_0 = 1. \quad (34)$$

The centered moment-generating function is

$$\mathcal{C}(T) := \mathbb{E}(e^{TZ}) = \sum_{k \geq 0} c_k \frac{T^{2k}}{(2k)!} \quad (35)$$

$$= \prod_{j=1}^{\infty} \frac{\sinh(T/2^j)}{T/2^j}. \quad (36)$$

The product is the real-exponential counterpart of the sinc product (2).

Define reciprocal centered moments γ_k by

$$\frac{1}{\mathcal{C}(T)} = \sum_{k \geq 0} \gamma_k \frac{T^{2k}}{(2k)!}. \quad (37)$$

Thus

$$\gamma_0 = 1, \quad \sum_{j=0}^r \binom{2r}{2j} \gamma_j c_{r-j} = 0 \quad (r \geq 1). \quad (38)$$

The first values are

$$\gamma_0 = 1, \quad \gamma_1 = -\frac{1}{9}, \quad \gamma_2 = \frac{31}{675}, \quad \gamma_3 = -\frac{2347}{59535}, \quad \gamma_4 = \frac{1904369}{32531625}. \quad (39)$$

Lemma 3.1 (Rationality and nonvanishing of the reciprocal moments). *Every γ_k is rational, and*

$$(-1)^k \gamma_k > 0. \quad (40)$$

In particular, $\gamma_k \neq 0$ for every k .

Proof. The logarithm of (36) is

$$\log \mathcal{C}(T) = \sum_{r \geq 1} \frac{2^{2r-1} B_{2r}}{r(2r)!(2^{2r} - 1)} T^{2r}, \quad (41)$$

so its coefficients, and hence those of its exponential reciprocal, are rational. For the sign, use

$$\frac{z}{\sinh z} = \sum_{r \geq 0} (-1)^r b_r z^{2r}, \quad b_r > 0. \quad (42)$$

For $r \geq 1$, this follows explicitly from

$$[z^{2r}] \frac{z}{\sinh z} = \frac{2(1 - 2^{2r-1}) B_{2r}}{(2r)!}. \quad (43)$$

Now

$$\frac{1}{\mathcal{C}(T)} = \prod_{j=1}^{\infty} \frac{T/2^j}{\sinh(T/2^j)}. \quad (44)$$

Every contribution to the coefficient of T^{2k} has the common sign $(-1)^k$, and at least one contribution is strictly positive in absolute value. This proves (40). $\square$

3.2 A dyadic grid point is the center of one polynomial cell

The knots of S_n are

$$\kappa_{r,n} = \frac{r + 1/2}{2^n}. \quad (45)$$

If $a = M/2^n$, then

$$[a - \delta_n, a + \delta_n] = \left[\frac{M - 1/2}{2^n}, \frac{M + 1/2}{2^n} \right] \quad (46)$$

is exactly one closed spline cell, centered at a . On its interior, S_n is a single polynomial of degree at most n . The exterior cells at $a = 0$ and $a = 1$ are constant and cause no exception.

Theorem 3.2 (Exact reciprocal moment deconvolution on a matched cell). *Let $n \geq 1$ and let $a = M/2^n \in [0, 1]$. Then*

$$S_n(a) = \sum_{k=0}^{\lfloor n/2 \rfloor} \frac{\gamma_k}{(2k)!} 4^{-k(n+1)} F^{(2k)}(a). \quad (47)$$

More generally, for $0 \leq r \leq n$, the n -jet satisfies the triangular moment relation

$$F^{(r)}(a) = \sum_{k=0}^{\lfloor (n-r)/2 \rfloor} \frac{c_k \delta_n^{2k}}{(2k)!} S_n^{(r+2k)}(a). \quad (48)$$

Proof. Let $P(h) = S_n(a + h)$ denote the polynomial branch on $[-\delta_n, \delta_n]$. Because $a - \delta_n Z$ remains in this cell almost surely, differentiating the compactly supported convolution (33) in the distributional sense, and then using the ordinary polynomial branch inside the cell, gives

$$F^{(r)}(a) = \mathbb{E}[P^{(r)}(-\delta_n Z)] \quad (0 \leq r \leq n). \quad (49)$$

The Taylor formula for the polynomial $P^{(r)}$ is finite. Odd moments of Z vanish because up is even, so taking expectations gives (48).

To invert the triangular relation, consider

$$T := \sum_{k=0}^{\lfloor n/2 \rfloor} \frac{\gamma_k \delta_n^{2k}}{(2k)!} F^{(2k)}(a). \quad (50)$$

Substitute (48). The coefficient of $\delta_n^{2r} S_n^{(2r)}(a)/(2r)!$ becomes

$$\sum_{k=0}^r \binom{2r}{2k} \gamma_k c_{r-k}. \quad (51)$$

By (38), this coefficient is 1 for $r = 0$ and 0 for every $r \geq 1$. Therefore $T = S_n(a)$. Finally, $\delta_n^{2k} = 2^{-2k(n+1)} = 4^{-k(n+1)}$, proving (47). $\square$

Remark 3.3 (Why the formula is exact). *There is no passage to an infinite Taylor series in this proof. At a matched dyadic point, the tail displacement $\delta_n Z$ stays inside one degree- n polynomial cell. Taylor expansion therefore terminates, and reciprocal moment deconvolution is finite algebra. This is the mechanism behind the eventual exact geometric behavior.*

4 Dyadic depth kills all but finitely many derivative modes

4.1 The signed global extension and derivative dilation

Introduce the signed global extension

$$\mathcal{G}(y) = \sum_{b \geq 0} \varepsilon_b \operatorname{up}(y - 2b - 1). \quad (52)$$

The sum is locally finite. On the first block,

$$\mathcal{G}(y) = \operatorname{up}(y - 1), \quad 0 \leq y \leq 2, \quad (53)$$

and on $[0, 1]$ it agrees with F . It satisfies the global differential refinement equation [4, 6]

$$\mathcal{G}'(y) = 2\mathcal{G}(2y). \quad (54)$$

Iterating gives

$$\mathcal{G}^{(r)}(y) = 2^{\binom{r+1}{2}} \mathcal{G}(2^r y). \quad (55)$$

In particular, for $a \in [0, 1]$,

$$F^{(2k)}(a) = 2^{k(2k+1)} \mathcal{G}(4^k a). \quad (56)$$

The block endpoints and centers give

$$\mathcal{G}(2b) = 0, \quad (57)$$

$$\mathcal{G}(2b + 1) = \varepsilon_b, \quad (58)$$

$$\mathcal{G}\left(2b + \frac{1}{2}\right) = \mathcal{G}\left(2b + \frac{3}{2}\right) = \frac{\varepsilon_b}{2}. \quad (59)$$

4.2 The finite derivative jet at a dyadic point

Let $a = m/2^s \in (0, 1)$ be reduced, so m is odd, and put $d = \lfloor s/2 \rfloor$.

Lemma 4.1 (Vanishing beyond half the dyadic depth). *For every $k > d$,*

$$F^{(2k)}(a) = 0. \quad (60)$$

For every $0 \leq k \leq d$, one has

$$F^{(2k)}(a) \neq 0. \quad (61)$$

Proof. If $s = 2d$, then $4^d a = m$ is an odd integer, while for every $k > d$, $4^k a = m4^{k-d}$ is an even integer. If $s = 2d + 1$, then $4^d a = m/2$ is an odd half-integer, while for $k > d$, $4^k a = 2m4^{k-d-1}$ is an even integer. Now use (56) and (59). At the top index, $\mathcal{G}(4^d a)$ is respectively ± 1 or $\pm 1/2$. For $k < d$, the number $4^k a = m/2^{s-2k}$ is not an integer, hence it lies in the interior of one of the two-unit blocks in (52); the up-function is strictly positive in the interior of its support, so $\mathcal{G}(4^k a) \neq 0$. $\square$

4.3 Proof of the exact eventual tail

For $n \geq s$, the reduced dyadic $a = m/2^s$ is also a point of the 2^{-n} grid:

$$a = \frac{m2^{n-s}}{2^n}. \quad (62)$$

Apply Theorem 3.2, then use Lemma 4.1. We obtain

$$q_n = S_n(a) = F(a) + \sum_{k=1}^d \frac{\gamma_k}{(2k)!} 4^{-k(n+1)} F^{(2k)}(a) \quad (63)$$

$$= \text{up}(x) + \sum_{k=1}^d A_k(x) 4^{-kn}, \quad (64)$$

where

$$A_k(x) = \frac{\gamma_k}{(2k)! 4^k} F^{(2k)}(a) = \frac{\gamma_k 2^{k(2k-1)}}{(2k)!} \mathcal{G}(4^k a), \quad a = 1 - |x|. \quad (65)$$

This proves the formula in Theorem 1.1. It also proves rationality without requiring a separate arithmetic theorem about F or $\mathcal{G}$. Indeed, each $q_n = S_n(a)$ is rational by the finite sum (16). Evaluating (64) at $n = s, s+1, \dots, s+d$ gives a square Vandermonde system over $\mathbb{Q}$ for $F(a), A_1(x), \dots, A_d(x)$, with distinct nodes $1, \rho, \dots, \rho^d$. Its unique solution is rational. In particular, this argument itself recovers the classical rationality of $\text{up}(x) = F(a)$ at dyadic arguments.

For $s \geq 2$, Lemma 3.1 and Lemma 4.1 imply $A_k(x) \neq 0$ for every $1 \leq k \leq d$. More explicitly, if $a = m/2^s$ is reduced, then

$$A_d(x) = \begin{cases} \frac{\gamma_d 2^{d(2d-1)}}{(2d)!} \varepsilon_{(m-1)/2}, & s = 2d, \\ \frac{\gamma_d 2^{d(2d-1)-1}}{(2d)!} \varepsilon_{\lfloor m/4 \rfloor}, & s = 2d + 1. \end{cases} \quad (66)$$

Thus the tail contains exactly d nonconstant geometric modes, all with nonzero coefficients.

Remark 4.2 (What happens before $n = s$). *For $n < s$, the point a is not a center of the 2^{-n} spline grid. The tail interval need not coincide with one polynomial cell, so the matched-cell deconvolution argument does not apply in this form. These are precisely the “few initial values” allowed in the question. Once $n \geq s$, the formula (64) is an identity for every subsequent n .*

5 Extraction from consecutive spline values

We now solve the purely algebraic problem created by Theorem 1.1. Let

$$q_n = L + \sum_{r=1}^d A_r \rho^{rn}, \quad \rho = \frac{1}{4}, \quad n \geq N. \quad (67)$$

In the application, $L = \text{up}(x)$ and any $N \geq s$ is valid.

5.1 The Vandermonde system

The $d + 1$ values $q_N, \dots, q_{N+d}$ satisfy

$$\begin{bmatrix} 1 & \rho^N & \rho^{2N} & \cdots & \rho^{dN} \\ 1 & \rho^{N+1} & \rho^{2(N+1)} & \cdots & \rho^{d(N+1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \rho^{N+d} & \rho^{2(N+d)} & \cdots & \rho^{d(N+d)} \end{bmatrix} \begin{bmatrix} L \\ A_1 \\ A_2 \\ \vdots \\ A_d \end{bmatrix} = \begin{bmatrix} q_N \\ q_{N+1} \\ \vdots \\ q_{N+d} \end{bmatrix}. \quad (68)$$

The determinant is nonzero because $1, \rho, \dots, \rho^d$ are distinct. Therefore all coefficients are uniquely determined over $\mathbb{Q}$.

A useful interpolation reformulation is

$$P_N(z) := L + \sum_{r=1}^d A_r \rho^{rN} z^r. \quad (69)$$

Then

$$P_N(\rho^j) = q_{N+j}, \quad 0 \leq j \leq d, \quad (70)$$

and the desired limit is simply

$$L = P_N(0). \quad (71)$$

Moreover, once P_N has been interpolated,

$$A_r = \rho^{-rN} [z^r] P_N(z). \quad (72)$$

Thus the same $d + 1$ values recover every coefficient, not only the limit. Explicitly,

$$P_N(z) = \sum_{j=0}^d q_{N+j} \prod_{\substack{0 \leq \ell \leq d \\ \ell \neq j}} \frac{z - \rho^\ell}{\rho^j - \rho^\ell}. \quad (73)$$

5.2 Lagrange weights at zero

Setting $z = 0$ in (73), the weight of q_{N+j} is

$$\lambda_{d,j}(\rho) := \prod_{\substack{0 \leq \ell \leq d \\ \ell \neq j}} \frac{-\rho^\ell}{\rho^j - \rho^\ell} \quad (74)$$

$$= (-1)^{d-j} \frac{\rho^{\binom{d-j+1}{2}}}{(\rho; \rho)_j (\rho; \rho)_{d-j}}. \quad (75)$$

Hence a forward-order formula is

$$L = \sum_{j=0}^d (-1)^{d-j} \frac{\rho^{\binom{d-j+1}{2}}}{(\rho; \rho)_j (\rho; \rho)_{d-j}} q_{N+j}. \quad (76)$$

Reindexing by $j \mapsto d - j$ and using the Gaussian binomial coefficient in (9) gives exactly

$$L = \frac{1}{(\rho; \rho)_d} \sum_{j=0}^d (-1)^j \rho^{\binom{j+1}{2}} \begin{bmatrix} d \\ j \end{bmatrix}_\rho q_{N+d-j}, \quad (77)$$

which proves Theorem 1.2.

5.3 The row polynomial and mode annihilation

Define the reverse-order weights

$$w_{d,j} = \frac{(-1)^j \rho^{\binom{j+1}{2}}}{(\rho; \rho)_d} \begin{bmatrix} d \\ j \end{bmatrix}_\rho, \quad 0 \leq j \leq d. \quad (78)$$

The finite q -binomial theorem [5] gives their generating polynomial:

$$W_d(z) := \sum_{j=0}^d w_{d,j} z^j \quad (79)$$

$$= \frac{(\rho z; \rho)_d}{(\rho; \rho)_d} = \prod_{r=1}^d \frac{1 - \rho^r z}{1 - \rho^r}. \quad (80)$$

Therefore

$$W_d(1) = 1, \quad W_d(\rho^{-r}) = 0 \quad (1 \leq r \leq d). \quad (81)$$

In the reverse combination $\sum_j w_{d,j} q_{N+d-j}$, the constant mode contributes $L W_d(1) = L$, while the mode $A_r \rho^{rn}$ contributes a multiple of $W_d(\rho^{-r}) = 0$. This is the shortest proof of the extraction formula once the geometric tail is known.

5.4 Shift-operator and triangular Richardson forms

Let E be the forward shift, $(Eq)_n = q_{n+1}$. Since E acts on the mode ρ^{rn} by the eigenvalue ρ^r , we have

$$L = \frac{\prod_{r=1}^d (E - \rho^r)}{\prod_{r=1}^d (1 - \rho^r)} q_N. \quad (82)$$

Expanding (82) is another route to (77).

It also gives an in-place triangular algorithm. Start with

$$R_0(j) = q_{N+j}, \quad 0 \leq j \leq d, \quad (83)$$

and, for $r = 1, \dots, d$, form

$$R_r(j) = \frac{R_{r-1}(j+1) - \rho^r R_{r-1}(j)}{1 - \rho^r}, \quad 0 \leq j \leq d - r. \quad (84)$$

Then

$$R_d(0) = L. \quad (85)$$

With rational input, every operation is rational.

5.5 Sample count and stability

For $s \geq 2$, the top coefficient A_d is nonzero. Within the model (67), there are $d + 1$ independent unknowns $L, A_1, \dots, A_d$, and the geometric Vandermonde matrix is nonsingular. Thus $d + 1$ samples are both sufficient and, without additional information, necessary to determine all coefficients.

For approximate rather than exact inputs, the extraction row is well conditioned. Since $0 < \rho < 1$ and the signs in (78) alternate,

$$\sum_{j=0}^d |w_{d,j}| = W_d(-1) = \frac{(-\rho; \rho)_d}{(\rho; \rho)_d}. \quad (86)$$

At $\rho = 1/4$, this increases to approximately

$$\frac{(-1/4; 1/4)_\infty}{(1/4; 1/4)_\infty} \approx 1.969260354. \quad (87)$$

Hence if every supplied q_n has absolute error at most η , the extracted constant has error at most about 1.97η , uniformly in the depth. Exact rational arithmetic, as used in the accompanying program, eliminates this issue entirely.

5.6 First extraction rows

For convenience, the first quarter-base rows are

$$d = 0: \quad L = q_N, \quad (88)$$

$$d = 1: \quad L = \frac{4}{3}q_{N+1} - \frac{1}{3}q_N, \quad (89)$$

$$d = 2: \quad L = \frac{64}{45}q_{N+2} - \frac{4}{9}q_{N+1} + \frac{1}{45}q_N, \quad (90)$$

$$d = 3: \quad L = \frac{4096}{2835}q_{N+3} - \frac{64}{135}q_{N+2} + \frac{4}{135}q_{N+1} - \frac{1}{2835}q_N. \quad (91)$$

6 Worked exact examples

6.1 Depth two: $x = \pm 3/4$

Here

$$a = 1 - |x| = \frac{1}{4}, \quad s = 2, \quad d = 1. \quad (92)$$

The exact dyadic value and derivative are

$$F\left(\frac{1}{4}\right) = \frac{5}{72}, \quad F''\left(\frac{1}{4}\right) = 8\mathcal{G}(1) = 8. \quad (93)$$

Since $\gamma_1 = -1/9$, formula (65) gives $A_1 = -1/9$. Therefore

$$\boxed{\text{up}_n\left(\pm\frac{3}{4}\right) = \frac{5}{72} - \frac{1}{9}4^{-n}, \quad n \geq 2.} \quad (94)$$

The first two values in the valid range are

$$q_2 = \frac{1}{16}, \quad q_3 = \frac{13}{192}. \quad (95)$$

The $d = 1$ extraction row gives

$$\frac{4}{3}q_3 - \frac{1}{3}q_2 = \frac{4q_3 - q_2}{3} = \frac{5}{72} = \text{up}\left(\pm\frac{3}{4}\right). \quad (96)$$

6.2 Depth four: $x = \pm 11/16$

Now

$$a = 1 - |x| = \frac{5}{16}, \quad s = 4, \quad d = 2. \quad (97)$$

The exact value is

$$L = F\left(\frac{5}{16}\right) = \frac{305857}{2073600}. \quad (98)$$

The derivative dilation gives

$$F''\left(\frac{5}{16}\right) = 8\mathcal{G}\left(\frac{5}{4}\right) = 8\text{up}\left(\frac{1}{4}\right) = 8F\left(\frac{3}{4}\right) = \frac{67}{9}, \quad (99)$$

$$F^{(4)}\left(\frac{5}{16}\right) = 2^{10}\mathcal{G}(5) = -1024. \quad (100)$$

Using $\gamma_1 = -1/9$ and $\gamma_2 = 31/675$ in (65),

$$A_1 = -\frac{67}{648}, \quad A_2 = -\frac{248}{2025}. \quad (101)$$

Thus the eventual formula is

$$\boxed{\text{up}_n\left(\pm \frac{11}{16}\right) = \frac{305857}{2073600} - \frac{67}{648} 4^{-n} - \frac{248}{2025} 16^{-n}, \quad n \geq 4.} \quad (102)$$

The exact spline values are

$$q_4 = \frac{1205}{8192}, \quad (103)$$

$$q_5 = \frac{289799}{1966080}, \quad (104)$$

$$q_6 = \frac{13917509}{94371840}. \quad (105)$$

Applying the $d = 2$ row,

$$\frac{64}{45}q_6 - \frac{4}{9}q_5 + \frac{1}{45}q_4 = \frac{305857}{2073600}, \quad (106)$$

exactly as claimed.

6.3 Depth zero and one

When $d = 0$, there is no decaying term. The spline sequence stabilizes as soon as the point lies on the grid. In particular,

$$\text{up}_n(0) = 1, \quad (107)$$

$$\text{up}_n(\pm 1) = 0, \quad (108)$$

$$\text{up}_n\left(\pm \frac{1}{2}\right) = \frac{1}{2} \quad (n \geq 1). \quad (109)$$

These are the $s = 0$ and $s = 1$ cases of the main theorem.

7 Exact computational method

For a given dyadic $x \in [-1, 1]$, the shortest exact workflow is:

1. Reduce $a = 1 - |x|$ to $m/2^s$ and set $d = \lfloor s/2 \rfloor$.

2. Evaluate $q_s, q_{s+1}, \dots, q_{s+d}$ from the finite spline formula (16). Every operation is rational.
3. Apply either the single q-binomial formula (8) or the triangular recursion (84). The result is exactly $\text{up}(x)$.
4. To recover the complete eventual formula, solve the rational Vandermonde system (68), or interpolate P_s using (73) and use (72).
5. Optionally verify the recovered coefficients against one or more later values. The theorem guarantees exact agreement for every $n \geq s$.

The included file `verify_extrapolation.py` implements all of these steps with Python's `Fraction` type. It contains:

- exact Thue–Morse spline evaluation;
- dyadic-depth detection;
- finite q-Pochhammer and Gaussian-binomial functions;
- both closed-row and recursive extraction;
- exact Gauss–Jordan solution of the full Vandermonde system;
- computation of reciprocal moments γ_k from Bernoulli numbers;
- assertions for the two examples above and checks at later n ;
- an optional exhaustive exact test over all reduced dyadics through a user-selected denominator depth.

A typical invocation is

```
python3 verify_extrapolation.py
python3 verify_extrapolation.py --x=-11/16
python3 verify_extrapolation.py --self-test-max-depth 7
```

The direct Thue–Morse sum has a cutoff of order $2^n a$, so it is intended as an exact verification tool at moderate depth. Faster evaluation can use the binary recurrences and formal arithmetic developed in the ProveIt sources; the extrapolation row itself is only quadratic-time in d in its recursive form.

8 Relation to the half-base q-row in the ProveIt development

The linked development contains a general half-base Gaussian-binomial row for shifted finite splines. Its row polynomial has the form

$$\frac{(z/2; 1/2)_n}{(1/2; 1/2)_n} \quad (110)$$

and annihilates modes 2^{-rn} . The result here is tailored to the specific centered prefix

$$\prod_{k=0}^n \text{sinc}\left(\frac{\pi t}{2^k}\right). \quad (111)$$

The centered tail law is even, so every odd moment vanishes. In the exact matched-cell deconvolution, only even derivative orders $2k$ survive, and the scale factor is

$$2^{-2k(n+1)} = 4^{-k(n+1)}. \quad (112)$$

After absorbing the fixed factor 4^{-k} into A_k , the sequence modes are 4^{-kn} . That is why the natural and minimal base in (8) is $\rho = 1/4$, not $1/2$. The quarter-base row uses only $\lfloor s/2 \rfloor + 1$ consecutive values, whereas a generic half-base filter that also prepares for absent odd modes would be longer.

There is also a notational distinction worth preserving: in parts of the linked source, a letter Q denotes a complex *translation parameter* in a shifted spline. The number $\rho = 1/4$ in this note is instead the *geometric decay ratio* of the centered finite-product sequence.

9 Conclusion

For a fixed dyadic input, the finite sinc-product approximations have a much stronger structure than ordinary convergence. Once the spline grid reaches the reduced denominator depth, the error is an exact finite combination of known geometric ratios:

$$\text{up}_n(x) - \text{up}(x) = A_1(x)4^{-n} + A_2(x)4^{-2n} + \cdots + A_{\lfloor s/2 \rfloor}(x)4^{-\lfloor s/2 \rfloor n}. \quad (113)$$

The analytic reason is the exact centering of the omitted random tail; the arithmetic reason is the vanishing of high even derivatives at dyadic points; and the algebraic reason the limit is extractable is the q-Vandermonde structure of the remaining modes.

The normalized q-Pochhammer polynomial

$$\frac{(\rho z; \rho)_d}{(\rho; \rho)_d} \quad (114)$$

packages the entire extraction row. Its value at 1 preserves the true constant, and its roots $\rho^{-1}, \dots, \rho^{-d}$ annihilate every decaying mode. Formula (8) is therefore both a closed proof and a practical exact algorithm for recovering the rational value $\text{up}(x)$ from finitely many rational spline approximations.

A A compact theorem-to-algorithm summary

Let $x \in [-1, 1]$ be dyadic.

$$\begin{aligned} a &= 1 - |x| = m/2^s \quad \text{in lowest terms,} \\ d &= \lfloor s/2 \rfloor, \quad \rho = 1/4, \\ q_n &= \text{up}_n(x) = S_n(a). \end{aligned}$$

Then, for every $n \geq s$,

$$\begin{aligned} q_n &= \text{up}(x) + \sum_{k=1}^d A_k(x) \rho^{kn}, \\ A_k(x) &= \frac{\gamma_k}{(2k)! 4^k} F^{(2k)}(a), \\ \frac{1}{\mathcal{C}(T)} &= \sum_{k \geq 0} \gamma_k \frac{T^{2k}}{(2k)!}, \\ \mathcal{C}(T) &= \prod_{j=1}^{\infty} \frac{\sinh(T/2^j)}{T/2^j}. \end{aligned}$$

Finally, for any $N \geq s$,

$$\text{up}(x) = \frac{1}{(1/4; 1/4)_d} \sum_{j=0}^d (-1)^j 4^{-\binom{j+1}{2}} \left[\begin{matrix} d \\ j \end{matrix} \right]_{1/4} \text{up}_{N+d-j}(x).$$

This one line is the requested extraction formula.

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